ANNEXURE-VIII

Syllabi of all courses of UG programmes
B. Tech.
V Semester
[Artificial Intelligence (AI)/ Information Technology (Artificial Intelligence and Robotics)/ Artificial Intelligence (AI) and Data Science/ Artificial Intelligence (AI) and Machine Learning]
for batch admitted in 2024-25

Centre for Artificial Intelligence

MACHINE LEARNING

(31243101)

COURSE OBJECTIVES

- To introduce fundamental concepts, mathematical foundations, and scope of machine learning.
- To develop skills for model evaluation, optimization, and practical application of ML algorithms.

UNIT-I: Introduction to ML: definition, applications, types of learning. ML pipeline: data loading, preprocessing, modeling, evaluation, deployment. Mathematical foundations: linear algebra basics, gradient descent, cost functions. Parametric vs. non-parametric models. Bias–variance trade-off. Python ML ecosystem: NumPy, Pandas, Scikit-learn.
UNIT-II: Linear models: Linear & Polynomial Regression, Ridge, Lasso, Logistic Regression. Tree-based methods: Decision Trees, Random Forest, Gradient Boosting. Instance-based: K-Nearest Neighbors. Probabilistic models: Naive Bayes, Gaussian Discriminant Analysis. Support Vector Machines (hard & soft margin, kernel trick).
UNIT-III: Clustering: K-means, hierarchical clustering, DBSCAN. Association rule mining: Apriori, FP-growth. Anomaly detection. Dimensionality reduction: PCA, SVD. Visualization of high-dimensional data (t-SNE basics).
UNIT-IV: Performance metrics: precision, recall, F1-score, confusion matrix, ROC-AUC. Cross-validation techniques. Hyperparameter tuning: Grid Search, Random Search, Bayesian Optimization. Ensemble methods: bagging, boosting (AdaBoost, Gradient Boosting, XGBoost basics). Overfitting/Underfitting prevention, regularization. Handling imbalanced datasets (SMOTE, resampling).
UNIT-V (DYNAMIC CONTENTS): Deep Learning overview: Neural networks, backpropagation, activation functions. Modern architectures: Transformers, Vision Transformers, CNN advancements.

RECOMMENDED BOOKS

1. Machine Learning: A Probabilistic Perspective, Kevin Murphy
2. Pattern Recognition and Machine Learning, Christopher Bishop
3. Introduction to Machine Learning with Python, Andreas C. Müller & Sarah Guido.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain the fundamental concepts, workflow, and mathematical foundations of machine learning.
CO2	Implement major supervised learning algorithms for regression and classification tasks.
CO3	Analyze unsupervised learning techniques to discover patterns in data.
CO4	Evaluate, optimize, and improve machine learning models using advanced techniques.
CO5	Design modern machine learning advancements.

	CO-PO Mapping Matrix													CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	3	2									1	3	2	2	2	2	2					
CO2	3	3	2		3							3	3	2	3	3	3	1	3			
CO3	2	3	2	2	3							3	3	2	3	2	2	1	3		1	
CO4	2	3	2	3	3			1	1	1		3	3	2	3	2	2	1	3		2	
CO5		2			2	1	1	1	1	1	3	3	3	1	1	1	1		2	1	2	1

Centre for Artificial Intelligence

THEORY OF COMPUTATION

(24243101)

COURSE OBJECTIVES

- To understand computability, decidability, and complexity through problem solving.
- To analyze and design abstract model of computation & formal languages
- To understand and conduct mathematical proofs for computation and algorithms.

UNIT-I: Introduction of Automata Theory: Examples of automata machines, Finite Automata as a language acceptor and translator, Moore machines and mealy machines, composite machine, Conversion from Mealy to Moore and vice versa.
UNIT-II: Types of Finite Automata: Non Deterministic Finite Automata (NDFA), Deterministic finite automata machines, conversion of NDFA to DFA, minimization of automata machines, regular expression, Arden's theorem. Meaning of union, intersection, concatenation and closure, 2 way DFA.
UNIT-III: Grammars: Types of grammar, context sensitive grammar, and context free grammar, regular grammar. Derivation trees, ambiguity in grammar, simplification of context free grammar, conversion of grammar to automata machine and vice versa, Chomsky hierarchy of grammar, killing null and unit productions. Chomsky normal form and Greibach normal form.
UNIT-IV: Push down Automata: example of PDA, deterministic and non-deterministic PDA, conversion of PDA into context free grammar and vice versa, CFG equivalent to PDA.
UNIT-V: Turing Machine: Techniques for construction. N-P complete problems. Decidability and Recursively Enumerable Languages, decidability, decidable languages, undecidable languages.

RECOMMENDED BOOKS

1. Introduction to Automata Theory Language & Computation, Hopcroft & Ullman, Narosa Publication.
2. Element of the Theory Computation, Lewis & Christors, Pearson.
3. Theory of Computation, Chandrasekhar & Mishra, PHI.
4. Theory of Computation, Wood, Harper & Row.
5. Introduction to Computing Theory, Daniel I-A Cohen, Wiley.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain the basic concepts of switching and finite automata theory & languages.
CO2	Classify practical problems to languages, automata, computability and complexity.
CO3	Construct abstract models of computing and check their power to recognize the languages.
CO4	Evaluate the grammar, its types, simplification and normal form.
CO5	Design formal mathematical methods to prove properties of languages, grammars and automata.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2		1								2	2	2	2	2	2					
CO2	3	3		2								2	2	2	2	3	3				1	
CO3	3	3	2	2	1							2	2	2	2	3	3	1	1		1	
CO4	2	3		3				1	1	1		2	2	2	2	2	2				2	
CO5	2	3		3		1	1	1	1	1	2	2	2	2	2	2	2			1	2	1

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DATA MINING AND WAREHOUSING

(27243101)

COURSE OBJECTIVES

- Study preprocessing and data transformation techniques
- Apply classification, clustering, and association algorithms
- Analyze real-world datasets using data mining methods

UNIT-I: Classification of data-Structured Data, Semi-Structured Data, Spatial Data, Time-Series Data, Transactional Data, need & significance of data mining, KDD, data mining functionalities, applications.
UNIT-II:Data Cleaning, Handling Missing Values, Noisy Data Removal, Data Integration, Data Transformation, Data Reduction, Data Discretization,Concept Hierarchies. Similarity Measures-Euclidean Distance, Manhattan Distance, Cosine Similarity. Association Rule Mining- Market Basket Analysis, Frequent Itemsets, Closed Frequent Itemsets, Algorithms- Apriori Algorithm, FP-Growth Algorithm, Association Rules-Support, Confidence, Lift, Correlation Analysis.
UNIT-III: Basics of Classification, Supervised Learning, Classification Process, Classification Techniques-Decision Tree Induction (Attribute Selection Measures, Information Gain, Gini Index, Tree Pruning), Bayes Theorem- Naïve Bayes Classifier, Bayesian Networks. Introduction to Clustering, Applications of Clustering, Types of Clustering, Clustering Methods- Partitioning Methods, Hierarchical Clustering, Density-Based Clustering, Grid-Based Clustering, K-Means Clustering, Advantages & Limitations.
UNIT-IV: Evolution & need of Data Warehousing, Operational Database Systems vs Data Warehouse,Characteristics- Subject-Oriented, Integrated, Time-Variant, Non-Volatile Properties, Architecture-Three-Tier Architecture, Enterprise Warehouse, Data Mart, Virtual Warehouse, Metadata Repository, Data Warehouse Design-Star Schema, Snowflake Schema,Fact Tables**,** OLTP vs OLAP, Data Cube, Roll-up, Drill-down, Slice and Dice, Pivoting, ROLAP, MOLAP.
UNIT-V: Introduction to Big data, characteristics, Big Data frameworks- Apache Hadoop, Spark, Hive, Pig, Machine Learning Integration- Model evaluation parameters, Modern algorithms like XGBoost, Random Forest, introduction to Deep Learning.

RECOMMENDED BOOKS

1.Data Mining: Concepts and Techniques, Han and Kamber, Morgan Kaufmann Publications
2.Data Mining Techniques, A. K. Pujari, Universities Press Pvt. Ltd.
3.Data Warehousing in the Real World, Sam Anahory, Pearson Publication.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain fundamental concepts of data types and data mining.
CO2	Classify data pre-processing steps and association rules for mining.
CO3	Compare data mining techniques for knowledge extraction from large amounts of data.
CO4	Evaluate methods for storing & retrieving data from different data sources.
CO5	Develop real world applications using data mining techniques.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2										3	2	2	2	2	2					
CO2	3	3			2							3	3	2	3	3	3		2			
CO3	2	3		2	2							3	3	2	2	2	2		2		1	
CO4	2	3	2		3			1	1	1		3	3	2	3	2	2	1	3			
CO5		2	3		3	1	1	1	1	1	2	3	3	1	2	1	1	2	3	1	1	1

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DEEP LEARNING

(28243101)

COURSE OBJECTIVES

- Understand CNNs, RNNs, LSTMs, and Transformer models for intelligent decision-making.
- Analyze the strengths and limitations of various deep learning architectures and techniques.
- Apply deep learning algorithms to solve real-world problems in computer vision, NLP, healthcare, and intelligent systems.

UNIT-I: **Machine Learning Basics:** Learning algorithms, Maximum Likelihood Estimation, Building Machine Learning Algorithms, Curse of Dimensionality, Introduction to Neural Networks, Multilayer Perceptron, Back-propagation Algorithm and its Variants, Gradient Descent Techniques: Batch Gradient Descent, Stochastic Gradient Descent, Mini-batch Gradient Descent, Optimization Techniques: Momentum, RMSProp, Adam Optimizer.

UNIT-II: **Deep Learning Architectures:** Machine Learning and Deep Learning, Representation Learning, Width and Depth of Neural Networks, Activation Functions: Sigmoid, Tanh, ReLU, Leaky ReLU (LReLU), ELU, Unsupervised Training of Neural Networks, Regularization Techniques: Dropout, Batch Normalization, Early Stopping, Hyperparameter Tuning.

UNIT-III: **Convolutional Neural Networks:** Architectural Overview, Motivation, Layers, Filters, Parameter Sharing, Pooling Operations, Regularization, Popular CNN Architectures: AlexNet, VGGNet, ResNet – Applications; Transfer Learning Techniques, Variants of CNN: DenseNet, PixelNet, Introduction to Vision Transformers (ViT).

UNIT-IV: **Sequence Modeling – Recurrent and Recursive NETS:** Recurrent Neural Networks, Bidirectional RNNs, Encoder–Decoder and Sequence-to-Sequence Architectures, Backpropagation Through Time (BPTT) for Training RNNs, Long Short-Term Memory Networks (LSTM), Gated Recurrent Unit (GRU), Attention Mechanism and Introduction to Transformer Models.

UNIT-V: Auto Encoders and Deep Generative Models – Undercomplete Autoencoder, Regularized Autoencoder, Stochastic Encoders and Decoders, Contractive Encoders, Variational Autoencoders (VAE); Deep Generative Models: Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines, Restricted Boltzmann Machines, Generative Adversarial Networks (GANs), Introduction to Diffusion Models, Autoencoders, Deep Learning Applications.

RECOMMENDED BOOKS

1. Deep Learning, MIT Press.
2. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, Aurélien Géron, O'Reilly Media.
3. Deep Learning with Python, François Chollet, Manning Publications.
4. Dive into Deep Learning, Cambridge University Press.
5. Generative Deep Learning, O'Reilly Media.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain the fundamentals of machine learning, neural networks, backpropagation, and optimization techniques.
CO2	Describe deep learning architectures, representation learning, activation functions, and regularization techniques.
CO3	Analyze Convolutional Neural Networks (CNNs), transfer learning techniques, and their applications in computer vision.
CO4	Evaluate sequence modelling using RNNs, LSTMs, GRUs, attention mechanisms, and Transformer models.
CO5	Design autoencoders and deep generative models such as GANs and RBMs to real-world deep learning applications.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	3	2									1	3	2	2	2	2	2					
CO2	3	2		2							2	3	3	2	2	2	2				2	
CO3	2	3		3	2						2	3	3	2	2	2	2		2		2	
CO4	2	3			2			1	1	1	2	3	3	2	2	2	2		2		1	
CO5			3	2	3	1	1	1	1	1	3	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence

CLOUD COMPUTING & VIRTUALIZATION

(31243102/ 27243102)

COURSE OBJECTIVES

- To understand Cloud Computing concepts, technologies, architecture and applications.
- To understand the underlying principle of cloud virtualization, cloud storage, data management and data visualization.
- To understand different cloud programming platforms and tools to develop and deploy applications on cloud.

UNIT-I: **Cloud Architecture and Model:** Technologies for Network-Based System, System Models for Distributed and Cloud Computing, NIST Cloud Computing Reference Architecture. Cloud Models:- Characteristics, Cloud Services, Cloud models (IaaS, PaaS, SaaS), Public vs Private Cloud, Cloud Solutions Cloud ecosystem, Service management, Computing on demand.

UNIT-II: **Virtualization:** Basics of Virtualization, Types of Virtualization, Implementation Levels of Virtualization, Virtualization Structures, Tools and Mechanisms, Virtualization of CPU, Memory, I/O Devices. Virtual Clusters and Resource management, Virtualization for Data-center Automation.Popular tools like VMWare, VirtualBox

UNIT-III: **Cloud Infrastructure:** Architectural Design of Compute and Storage Clouds, Layered Cloud Architecture Development, Design Challenges, Inter Cloud Resource Management, Resource Provisioning and Platform Deployment, Global Exchange of Cloud Resources.

UNIT-IV: **Programming Model:** Parallel and Distributed Programming Paradigms- MapReduce, Twister and Iterative MapReduce, Hadoop Library from Apache, Google App Engine (GAE), Amazon Web Service (AWS), Smart Cloud, Public Clouds and Service Offerings, Microsoft Windows Azure.
Mapping Applications, Programming Support, Cloud Software Environments, Eucalyptus, Open Nebula, OpenStack, Aneka, CloudSim.

UNIT-V: Security in the Cloud: Security Overview, Cloud Security Challenges and Risks, Software-as-a-Service Security, Security Governance, Risk Management, Security Monitoring, Security Architecture Design, Data Security, Application Security, Virtual Machine Security, Identity,Management and Access Control, Docker containers and Kubernetes.

RECOMMENDED BOOKS

1. Kai Hwang, Geoffrey C Fox, Jack G Dongarra, "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers, 2012.
2. John W. Rittinghouse and James F.Ransome, "Cloud Computing: Implementation, Management, and Security", CRC Press, 2010.
3. Toby Velte, Anthony Velte, Robert Elsenpeter, "Cloud Computing, A Practical Approach", TMH, 2009.
4. Kumar Saurabh, "Cloud Computing — insights into New-Era Infrastructure", Wiley India,2011
5. George Reese, "Cloud Application Architectures: Building Applications and Infrastructure in the Cloud" O'Reilly
6. James E. Smith, Ravi Nair, "Virtual Machines: Versatile Platforms for Systems and Processes", Elsevier/Morgan Kaufmann, 2005.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Describe cloud computing fundamentals.
CO2	Classify architecture, infrastructure and delivery models.
CO3	Apply suitable virtualization concepts based on problem characteristics.
CO4	Choose appropriate programming models for a specific cloud computing application.
CO5	Evaluate various security issues in cloud computing to create an efficient cloud computing environment.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1	1	2	1			1	2	1	1	1	2	2	2	2	1	2	1	1	
CO2	3	3	2	1	2		1		1	1	1	2	2	2	3	3	3	1	2		1	1
CO3	3	3	3	2	3	1			2	1	2	3	3	2	3	3	3	2	3	1	2	
CO4	3	3	3	3	3			1	1	1	2	3	3	2	3	3	3	2	3		2	
CO5	3	3	2	2	3	3	1	2	1	2	1	2	3	3	3	3	3	2	3	3	2	1

Centre for Artificial Intelligence
INTERNET OF THINGS
(247243102)

COURSE OBJECTIVES:

- Understand the fundamentals and architecture of IoT systems.
- Apply cloud, edge computing, security, and IoT analytics concepts.
- Design smart IoT applications for real-world domains.

UNIT–I : Introduction and IoT Architectures – Introduction to Internet of Things, evolution, characteristics, importance; Basic elements of IoT: sensors, actuators, gateways, and cloud platforms; IoT architectures: three-layer, five-layer, and OSI-based models; Communication models: device-to-device, device-to-cloud, and gateway-based communication; Design constraints, scalability, privacy, and security challenges in IoT systems.

UNIT–II : IoT Communication Networks and Technologies – IoT communication networks and Machine-to-Machine (M2M) communication. Network topologies, Short-range communication technologies: Bluetooth Low Energy (BLE), Zigbee, and IEEE 802.15.4; LPWAN technologies: LoRaWAN, LTE-M, and NB-IoT. IoT communication protocols at data link, network, and application layers; Standardization, interoperability, interference issues, deployment challenges, applications.

UNIT–III : Controllers and Protocol Stack using Sensors and Actuators – Study of multiple IoT microcontrollers and development boards: Arduino Nano, Arduino Uno, ESP32, NodeMCU, Raspberry Pi. IoT protocol stack concepts: 6LoWPAN, IPv6 addressing. Applications of sensors and actuators in smart homes, healthcare, industrial automation, agriculture, and environmental monitoring.

UNIT–IV : Cloud Computing, Messaging and Security – Introduction to cloud computing, edge computing in Internet of Things. Messaging protocols: CoAP, MQTT, AMQP; Client-server and publish-subscribe architectures, data formats, IoT data analytics, cloud-based monitoring systems; IoT security principles and protocols: Transport Layer Security (TLS), DTLS, and LwM2M.

UNIT–V : IoT Applications and Case Studies – Application of IoT business models, deployment strategies, and real-world case studies. Development of innovative IoT applications in smart homes, healthcare, industrial automation, agriculture, robotics, drone technology, and smart transportation.

RECOMMENDED BOOKS

1. Internet of Things: Architecture and Design, Morgan Kaufmann.
2. Internet of Things: A Hands-On-Approach, VPT.
3. The Internet of Things: Key Applications and Protocols, Wiley.
4. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things, Cisco Press.
5. Protocols and Architectures for Wireless Sensor Networks, Wiley.
6. Internet of Things: Principles and Paradigm, Morgan Kaufmann.
7. Internet of Things: Architecture and Design Principles, McGraw Hill Education.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Describe IoT fundamentals, architectures, and design challenges.
CO2	Analyze IoT communication networks and protocols.
CO3	Classify sensors, actuators, microcontrollers, and IoT protocol stacks.
CO4	Evaluate cloud computing, messaging, security, and IoT data analytics concepts.
CO5	Design and develop innovative IoT applications for real-world domains.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1		1							2	2	2	2	2	2		1			
CO2	3	3	2	2	3							2	2	2	3	3	3	1	3		1	
CO3	2	3	3	2	3							2	2	2	3	2	2	2	3		1	
CO4	2	3	3	3	3	1		1	1	1		2	2	2	3	2	2	2	3	1	2	
CO5	2	2	3	3	3	2	2	1	2	2	1	3	3	2	2	2	2	2	3	2	2	2

Centre for Artificial Intelligence
CLOUD COMPUTING & BIG DATA
(28243102)

COURSE OBJECTIVES

- To understand cloud computing concepts, architectures and service models.
- To study virtualization, cloud deployment models and cloud services.
- To understand big data technologies and analytics techniques.
- To develop practical skills in cloud platforms and big data processing tools.

UNIT-I: Introduction to Cloud Computing
Cloud computing concepts, characteristics, service models (IaaS, PaaS, SaaS), deployment models, cloud architecture, cloud providers.

UNIT-II: Virtualization and Cloud Services
Virtualization concepts, hypervisors, virtual machines, storage virtualization, cloud infrastructure, resource management.

UNIT-III: Big Data Fundamentals
Introduction to big data, characteristics of big data, big data lifecycle, distributed systems, Hadoop ecosystem.

UNIT-IV: Big Data Processing Frameworks
HDFS, MapReduce, YARN, Spark framework, data analytics and processing techniques.

UNIT-V: Cloud Security and Applications
Cloud security issues, privacy challenges, data protection, cloud applications, IoT and AI integration.

RECOMMENDED BOOKS

1. Rajkumar Buyya, Cloud Computing: Principles and Paradigms.
2. Tom White, Hadoop: The Definitive Guide.
3. Judith Hurwitz, Big Data for Dummies.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain cloud computing models and architecture.
CO2	Analyze big data technologies and distributed systems
CO3	Analyze restoration and morphological processing methods.
CO4	Implement Hadoop and big data processing frameworks.
CO5	Apply cloud and big data solutions in real-world applications.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1		2	1				1			3	2	2	2	2	1	2	1		
CO2	2	3	3	2	3	1	1						3	2	3	2	2	2	3	1	1	1
CO3	1	3	2	1	2	3							2	2	2	2	2	2	2	3		
CO4	1	2	3	3	2			1	1	1		1	3	1	2	2	2	2	2		2	
CO5	1	2	3	1	2	3	1	3	3	2	1	1	2	2	2	2	2	3	2	3	1	1

Centre for Artificial Intelligence

EVOLUTIONARY ALGORITHMS AND OPTIMIZATION

(31243103/ 27243103/ 28243103)

COURSE OBJECTIVES

- To understand the principles of evolutionary algorithms and optimization techniques inspired by natural evolution.
- To develop the capability to formulate and solve optimization problems using evolutionary algorithms.

UNIT I: Foundation of Optimization and Evolutionary Algorithms (EA) – Introduction to Optimization, Search space representation, Objective function and fitness landscape, Exploration vs exploitation trade-off, Generalized Formulation, Scope of Optimization via Applications, Characteristic of Optimization Functions; Principles of EA: Natural Evolutional and Genetics, Generalized Framework, Behaviour and Typical run of EA.

UNIT II: Genetic Algorithms and Differential Evolution – Genetic Algorithms (GA): principles and workflow, Chromosome encoding techniques, Selection methods: Roulette Wheel, Tournament Selection, Crossover operators and mutation operators, Elitism and replacement strategies, Constraint handling in GA, Differential Evolution (DE): Introduction, Concepts, Operators, Algorithm, DE on Generalized Framework, Performance analysis and parameter tuning, Simulated Annealing.

UNIT III: Swarm Intelligence and Nature-Inspired Optimization – Introduction to swarm intelligence, Velocity and position update equations, Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Artificial Bee Colony (ABC) Optimization, Firefly Algorithm and Bat Algorithm, Comparative analysis of swarm-based algorithms, Applications in routing, clustering, and feature selection

UNIT IV: Multi-objective and Hybrid Optimization – Introduction to Multi-Objective Optimization: Introduction, Generalized Formulation, Concept of Dominance and Pareto-optimality, Terminologies, Difference with Single-objective optimization, Approaches to multi-objective optimization; Classical Multi-Objective Optimization Methods: Weighted- Sum Method, ε-Constraint Method, Weighted Metric Methods, Hand Calculations, Difficulties with Classical approaches, Ideal Multi- Objective Optimization Approach

UNIT V: Multi-Objective Evolutionary Algorithms (MOEAs) – Introduction, MOEAs on generalized Framework, Algorithms: NSGA-II, SPEA2, Evolutionary Deep Learning, Neuro-evolution, AutoML and Neural Architecture Search, Evolutionary Reinforcement Learning, Quantum-inspired Optimization, AI-based Industrial Optimization, Applications in Robotics, Healthcare, Smart Cities, and Cybersecurity

TEXT & REFERENCE BOOKS

1. *Introduction to Evolutionary Computing* – A. E. Eiben and J. E. Smith, Springer.
2. *Genetic Algorithms in Search, Optimization and Machine Learning* – David E. Goldberg.
3. *Multi-Objective Optimization Using Evolutionary Algorithms* – Kalyanmoy Deb.
4. *Practical Genetic Algorithms* – Randy L. Haupt and Sue Ellen Haupt.
5. *Swarm Intelligence* – James Kennedy and Russell Eberhart.
6. *Essentials of Metaheuristics* – Sean Luke.
7. *Hands-On Genetic Algorithms with Python* – Eyal Wirsansky.

COURSE OUTCOMES (COs)

After successful completion of the course, students will be able to:

CO1	Explain the fundamentals of optimization and evolutionary computation techniques.
CO2	Apply Genetic Algorithms and swarm-based approaches to solve optimization problems.
CO3	Analyze and compare single-objective and multi-objective optimization techniques.
CO4	Evaluate constraint handling and evolutionary optimization methods for real-world applications.
CO5	Design modern advancements and research trends in evolutionary optimization.

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CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	3	2	1	2						2	2	2	2	3	3	3	1	2		2	
CO2	3	3	3	2	3				1		2	3	3	2	3	3	3	2	3		2	
CO3	3	3	2	2	3						2	2	3	2	3	3	3	1	3		2	
CO4	3	3	3	3	3	1	1	1	1	1	2	3	3	2	3	3	3	2	3	1	2	1
CO5	2	2	3	2	3	1	2	1	1	1	3	3	3	2	2	2	2	2	3	1	2	2

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ROBOT OPERATING SYSTEM

(24243103)

COURSE OBJECTIVES

- To study Robot programming fundamentals and ROS platform
- To understand the installation and applications of ROS
- To understand Robot Navigation through Simulations.

UNIT-I: Basics of Ubuntu/Linux: Linux basic commands for robotics, Linux IDE & Text Editor: VS Code, nano, gedit, Linux and I/O board communication: USB to Serial communication, Linux and Camera interfacing, Linux kernel customization, ROS2 and Linux commands, GPIO handling on Raspberry Pi.

UNIT-II: **ROS Architecture and concepts**: ROS Installation, Equation, Environment configuration, Nodes, Topic, services, parameters, actions and rosbags, Workspace and package, ros launch files and launching the nodes, integrating the launch file in ROS2 packages through substitution and event handler.

UNIT-III: **Learning ROS programming using TurtleSim**: ROS concepts using TurtleSim, Rviz, rqt graphs and Client libraries roscpp and rospy, publisher and subscriber communication model, service and client model, creating custom msg and srv files; Understanding Transformation and frames, Working with TF broadcaster and listener, Creating TF for your robot, Working with ROS TF tools.

UNIT-IV: **Learning Basics of Robot Simulator**: Getting started with Robot simulator concepts, Robot simulators: TurtleSim, Webots, Gazebo and MSsim; Robot Simulation Programming: Unified Robot Description format (URDF): Building a visual and movable Robot model; Adding physical and collision properties, Generation; Xacro, URDF with robot state publisher.

UNIT-V: Applications of ROS Simulators in robot manipulators and Robot Navigations, Moveit, Study of ROS2 control in differential and three-wheel drive in automotive vehicles, Drones, UAVs.

RECOMMENDED BOOKS

1. Robot Operating System for Absolute Beginners: Robotics Programming Made Easy.
2. “Programming Robots with ROS” by Quigley, Gerkey and Smart.
3. “The Linux Command Line” by William Shotts.
4. “It-Yourself Guide to the Robot Operating System: Volumes” by Patrick Goebel.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Describe the dedicated operating system for robots.
CO2	Explain the fundamental Requirements of ROS.
CO3	Illustrate robot Architecture and its concepts.
CO4	Analyze various Robot simulators and its programming.
CO5	Develop the working principle of various ROS in various Robotic applications.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1	1	2				1	1		2	3	2	2	2	2		2			
CO2	3	2	2	1	3				1	1		2	3	2	3	2	2	1	3			
CO3	3	3	2	2	3	1			1	1		2	3	2	3	3	3	2	3	1	1	
CO4	2	2	3	2	3			1	1	1	1	2	2	1	2	2	2	2	3		2	
CO5	3	3	3	2	3	1	1	1	2	1	1	3	3	2	3	3	3	2	3	1	2	1

Centre for Artificial Intelligence

STATISTICAL COMPUTING

(31243104)

COURSE OBJECTIVES

- Understand how computational methods support modern statistical analysis.
- Apply programming techniques to automate statistical workflows.
- Develop practical skills for data preprocessing, simulation, estimation, and visualization.

UNIT-I: Computational Data Handling and Exploratory Analytics – Introduction to statistical computing environments; Computational workflow for statistical analysis; Structured and unstructured data representation: Arrays, matrices, tables, and data frames; Dataset importing, cleaning, and preprocessing; Automated handling of missing values, duplicate records, inconsistent data types; Computational exploratory data analysis (EDA); Outlier detection using statistical and distance-based methods; Feature scaling and normalization; Basic dimensionality reduction intuition; Data preprocessing pipelines for AI applications.

UNIT-II: Statistical Programming and Computational Thinking – Functional programming concepts for statistics; Modular statistical programming; Efficient matrix and array computation; Vectorized statistical operations; Computational efficiency and memory considerations; Searching and sorting in statistical datasets; Recursive and iterative computational approaches; Building reusable statistical functions; Simulation-based computational workflows; Debugging and validation of statistical programs

UNIT-III: Computational Inference and Statistical Visualization – Computational implementation of inferential statistics; Programmatic computation of: confidence intervals, test statistics, p-values; Computational framework for: t-tests, chi-square tests, correlation analysis; Introduction to simulation-based inference; Resampling intuition for statistical validation; Statistical visualization techniques: distribution plots, pair plots, heatmaps, regression plots, time-series visualization; Visual interpretation of statistical patterns; Dashboard-style statistical reporting.

UNIT-IV: Computational Estimation and Predictive Modeling – Computational parameter estimation concepts; Numerical implementation of: Method of Moments, Maximum Likelihood Estimation; Introduction to optimization-based estimation; Gradient descent for parameter learning; Loss functions and error minimization; Linear and polynomial regression using computation; Residual analysis and model diagnostics; Training-testing data split; Cross-validation techniques; Bias-variance intuition; Introduction to regularization concepts; Computational model comparison using AIC and BIC.

UNIT-V: Simulation, Resampling, and Scalable Statistical Computing – Pseudo-random number generation, Simulation of statistical distributions, Monte Carlo simulation methods, Computational study of sampling variability, Bootstrapping for confidence estimation, Permutation testing for significance analysis, Jackknife resampling intuition, Randomized algorithms for statistical analysis, Introduction to computational complexity in statistics, Memory-efficient data processing concepts, Introduction to scalable statistical workflows, Applications in AI, forecasting, and decision systems

RECOMMENDED BOOKS

1. Givens, G. H., and Hoeting, J. A. Computational Statistics. John Wiley & Sons.
2. James, G., Witten, D., Hastie, T., and Tibshirani, R. An Introduction to Statistical Learning. Springer.
3. Efron, B., and Hastie, T. Computer Age Statistical Inference: Algorithms, Evidence, and Data Science. Cambridge University Press.
4. Bruce, P., Bruce, A., and Gedeck, P. Practical Statistics for Data Scientists: 50+ Essential Concepts. O'Reilly Media.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain computational workflows for preprocessing, analyzing, and managing statistical datasets.
CO2	Develop modular statistical programs using vectorized and efficient computational approaches.
CO3	Analyze computational statistical inference and create meaningful statistical visualizations.

CO4	Evaluate computational estimation and model validation techniques for predictive analytics.
CO5	Design simulation-based statistical analysis and resampling-driven inference for practical AI applications.

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CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2	3		2	3							2	3	2	3	2	2		3		1	
CO2	2	3	2		3							2	3	2	3	2	2	1	3			
CO3	2	3		3	3							2	3	2	3	2	2		3		2	
CO4	2	3	2	3	3			1	1	1		3	3	2	3	2	2	1	3		2	
CO5			2	2	3	1	1	1	1	1	3	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence

DATA SCIENCE

(24243104)

COURSE OBJECTIVES

- To provide foundational understanding of data science concepts, workflow, and applications.
- To equip students with essential skills in data collection, cleaning, preprocessing, statistical analysis, core machine learning algorithms and exploratory data analysis.

UNIT-I: Data Science overview, roles & responsibilities of a data scientist, CRISP-DM methodology, Types of data, data collection methods, data quality issues. Data acquisition: APIs, web scraping, databases. Data cleaning: handling missing values, outliers, duplicates.
UNIT-II: Data transformation: normalization, standardization, encoding categorical variables, feature scaling. Feature engineering: extraction, selection, dimensionality reduction (PCA basics). Visualization using Matplotlib, Seaborn, Plotly. Advanced visualizations: heatmaps, pair plots, box plots, violin plots.
UNIT-III: Descriptive statistics: mean, median, variance, skewness. Probability distribution functions: Normal, Binomial, Poisson, Exponential. Hypothesis testing: z-test, t-test, chi-square test. Correlation and covariance. Regression basics: simple and multiple linear regression; evaluation metrics.
UNIT-IV: Types of machine learning: supervised, unsupervised, reinforcement learning. Supervised learning: Linear regression, Logistic regression, KNN, Naive Bayes, Decision Trees, Random Forest, SVM. Unsupervised learning: K-Means clustering, Hierarchical clustering, DBSCAN, PCA. Model evaluation metrics (Accuracy, Precision, Recall, F1, ROC-AUC, Confusion matrix, RMSE, MAE). Overfitting, underfitting, bias-variance tradeoff, cross-validation, regularization.
UNIT-V: Big Data technologies overview (Hadoop, Spark basics). Deep Learning fundamentals and CNN/RNN concepts, frameworks (TensorFlow/PyTorch).

RECOMMENDED BOOKS

1. Python Data Science Handbook, Jake VanderPlas, O'Reilly Media, 2016.
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Aurélien Géron, 3rd Edition, O'Reilly Media, 2022.
3. An Introduction to Statistical Learning with Applications in R (or Python), Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, 2nd Edition, 2021.
4. Python for Data Analysis, Wes McKinney, 3rd Edition, O'Reilly, 2022.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain the fundamental concepts, workflow, and tools of data science.
CO2	Apply appropriate data preprocessing and visualization techniques.
CO3	Analyze statistical reasoning and hypothesis testing for data-driven decisions.
CO4	Evaluate various supervised and unsupervised machine learning algorithms.
CO5	Design emerging trends and advanced technologies in Data Science.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2			1							3	2	2	2	2	2		1			
CO2	2	3			2							2	3	2	2	2	2		2			
CO3	2	3		2								2	3	2	2	2	2				1	
CO4	2	3	2		3			1	1	1		3	3	2	3	2	2	1	3			
CO5					2	1	1	1	1	1	3	2	3		1				2	1	2	1

Centre for Artificial Intelligence

IMAGE PROCESSING

(28243104)

COURSE OBJECTIVES

- To understand fundamentals of digital image processing and image representation techniques.
- To develop analytical and implementation skills for image processing algorithms.

UNIT-I: Introduction to Digital Image Processing – Introduction, applications of image processing, image sensing and acquisition, sampling and quantization, image representation, pixel relationships, color models, basic image transformations.
UNIT-II: Image Enhancement Techniques – Spatial domain processing, gray level transformations, histogram processing, smoothing filters, sharpening filters, frequency domain enhancement techniques.
UNIT-III: Image Restoration and Morphological Processing – Image degradation models, noise models, restoration filters, inverse filtering, Wiener filtering, morphological operations, dilation, erosion, opening and closing.
UNIT-IV: Image Segmentation and Representation – Thresholding techniques, edge detection, region-based segmentation, image representation methods, boundary descriptors and feature extraction.
UNIT-V: Image Compression and Applications – Image compression techniques, lossless and lossy compression, JPEG compression, applications in computer vision, medical imaging and AI-based image processing.

RECOMMENDED BOOKS

1. Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Pearson.
2. Anil K. Jain, Fundamentals of Digital Image Processing, PHI.
3. S. Sridhar, Digital Image Processing, Oxford University Press.
4. William K. Pratt, Digital Image Processing, Wiley.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain fundamentals and representation techniques of digital images.
CO2	Apply image enhancement and filtering techniques.
CO3	Implement distributed computing frameworks and MapReduce programming models for large-scale data processing.
CO4	Evaluate segmentation and feature extraction techniques.
CO5	Apply image compression techniques.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2			1							3	2	2	2	2	2		1			
CO2	3	3	2	2	3							3	3	2	3	3	3	1	3		1	
CO3	3	3	2	3	2							3	2	2	3	3	3	1	2		2	
CO4	3	3	3	3	3			1	1	1		3	3	2	3	3	3	2	3		2	
CO5	2	3	3	2	3	1	1	1	1	1	1	2	2	2	3	2	2	2	3	1	2	1

Centre for Artificial Intelligence

SUPPLY CHAIN MANAGEMENT

(31243110/ 24243110/ 27243110/ 28243110)

COURSE OBJECTIVES

- To introduce foundational concepts of supply chain networks and traditional logistics frameworks.
- To explore applications of machine learning, computer vision and robotics in supply chain management.

UNIT-I: Introduction to SCM and AI Integration – SCM Core Components: Drivers of performance; facilities, inventory, and transportation. Traditional Inefficiencies: Information asymmetry and the Bullwhip Effect. AI Transformation: Shift from reactive to predictive operations using Big Data. Data Pipelines: Sourcing data from IoT sensors, ERP systems, and external APIs.

UNIT-II: Predictive Demand and Inventory Optimization – Demand Forecasting: Traditional time-series models vs. Machine Learning (XGBoost, LSTM). Feature Engineering: Incorporating seasonality, pricing trends, and market indicators. Inventory Control: AI-driven Multi-Echelon Inventory Optimization (MEIO). Automated Replenishment: Dynamic safety stock calculations using probabilistic ML models.

UNIT-III: Logistics and Routing Optimization – Network Design: Facility location problem solving using K-Means and Linear Programming. Vehicle Routing Problem (VRP): Classic VRP, Capacitated VRP, and Time Window constraints. AI Solvers: Genetic Algorithms and Reinforcement Learning for dynamic, traffic-aware routing. Last-Mile Analytics: Geospatial analytics for delivery window prediction.

UNIT-IV: Smart Warehousing and Intelligent Procurement – Spatial AI: Slotting optimization and automated layout planning. Computer Vision: Object detection (YOLO) for automated stock counting and damage inspection. Warehouse Robotics: Pathfinding algorithms (A*, D*) for Autonomous Mobile Robots (AMRs). Procurement Analytics: Natural Language Processing (NLP) for automated contract review.

UNIT-V: Risk Management and Sustainability – Predictive Risk: Sentiment analysis of news and weather data to forecast port disruptions. Supplier Selection: Multi-criteria decision-making using AI-driven vendor scoring. Traceability: Combining AI and Blockchain for provenance and fraud detection. Green Supply Chains: AI frameworks for carbon footprint and reverse logistics optimization.

RECOMMENDED BOOKS

1. Supply Chain Management: Strategy, Planning, and Operation, by Sunil Chopra and Peter Meindl, Pearson.
2. Data Science for Supply Chain Forecasting, by Nicolas Vandeput, De Gruyter.
3. Deep Learning for Logistics and Supply Chain Management, by Jan Holmström et al., Springer.
4. Introduction to Computational Optimization Models for Logistics, by Alan L. Erera, CRC Press.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain supply chain structures and operations suitable for AI intervention.
CO2	Illustrate regression and time-series ML models to predict market demand.
CO3	Evaluate logistics networks and optimize vehicle routing.
CO4	Assess warehouse efficiency using computer vision and automated pathfinding workflows.
CO5	Formulate a resilient supply chain strategy leveraging predictive risk analytics.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2			1	1				1	2	1	1	2	2	3	3	1	1	1	1	
CO2	3	3	2	2	3				1	2	2	2	2	2	3	3	3	1	3		2	
CO3	3	3	3	2	3	1				2	1	2	2	2	3	3	3	2	3	1	2	
CO4	2	3	3	3	3	2				2	2	3	3	2	3	3	3	3	3	2	3	
CO5	2	2	3	2	2	3	2	1	1	3	3	2	2	2	2	2	2	3	2	3	3	2

ANNEXURE-IX

Experiment list/ Lab manual and project list (Cornerstone Project) of all the Laboratory Courses B.Tech.
V Semester
[Artificial Intelligence (AI)/ Information Technology (Artificial Intelligence and Robotics)/ Artificial Intelligence (AI) and Data Science/ Artificial Intelligence (AI) and Machine Learning]
for batch admitted in 2024-25

Centre for Artificial Intelligence

MACHINE LEARNING LAB

(31243105)

1. Introduction to Python for ML: Basics of Python, NumPy, Pandas, Matplotlib, Seaborn.
2. Data Preprocessing: Handling missing values, outlier detection. Encoding (label, one-hot), feature scaling (standardization/normalization).
3. Exploratory Data Analysis (EDA): Descriptive statistics, distribution plots, correlation heatmaps.
4. Train-Test Split and Cross-Validation: Dataset partitioning and performing k-fold cross-validation.
5. Linear Regression: Simple & Multiple linear regression. Model evaluation: RMSE, MAE, R^2 score.
6. Logistic Regression: Binary classification using logistic regression. Confusion matrix, ROC curve, AUC.
7. K-Nearest Neighbors (KNN): Classification using distance metrics. Optimal K selection using cross-validation.
8. Naïve Bayes Classifier: Gaussian/Multinomial NB on text or tabular data. Spam detection / sentiment classification.
9. Support Vector Machines (SVM): Linear and RBF kernel. Hyperparameter tuning using GridSearchCV.
10. Decision Trees: Classification & regression trees. Visualizing the tree. Impact of max_depth, min_samples_split.
11. Random Forests & Ensemble Methods: Bagging, Random Forest, and feature importance. Comparing single tree vs ensemble performance.
12. K-Means Clustering: Clustering workflow. Elbow method and silhouette score.
13. Hierarchical Clustering: Agglomerative clustering. Dendrogram visualization.
14. Dimensionality Reduction (PCA): Eigenvalues/eigenvectors, explained variance ratio. Visualization in 2D space.
15. Artificial Neural Networks (ANN) – Basic: Building a simple feedforward network using TensorFlow/Keras or Scikit-learn. Training, validation, accuracy evaluation.
16. Model Evaluation & Hyperparameter Tuning: GridSearchCV, RandomizedSearchCV. Bias-variance tradeoff.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Apply data preprocessing, cleaning, and preparation techniques essential for machine learning tasks.
CO2	Implement supervised & unsupervised learning algorithms and evaluate them using appropriate performance metrics.
CO3	Analyze an end-to-end machine learning workflow.

CO-PO Mapping Matrix														**CO-WK Mapping Matrix**								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2		3	2	3							3	3	1	2	1	1	2	3		1	
CO2	2	3	3	2	3							3	3	2	3	2	2	2	3		1	
CO3		2	3	3	3			1	1	1	2	3	3	1	2	1	1	2	3		2	
CO4	2	2			2	1	1					2	2	2	2	2	2		2	1		1
CO5		2			3			1	1	1	2	2	3	1	2	1	1		3		1	

Centre for Artificial Intelligence

ROBOT OPERATING SYSTEM LAB

(24243105)

1. To install the Linux Operating System using Dual Boot using Virtual Box.
2. Introduction to Ubuntu/Linux Commands for Robotics.
3. To study the Installation of Robot operating System and Configuration of ROS2 Environment
4. To create Creation of Workspace and Packages for Robot operating system.
5. To study the Nodes creation using Turtle Sim and its teleoperation process
6. Develop ROS2 publisher and subscriber nodes to control TurtleSim robot movement.
7. Implement service-client model in ROS2 for robot command execution and response handling.
8. Visualize robot data, node communication, TF frames, and topic connections using RViz and rqt tools.
9. Create a basic robot model using URDF and visualize the robot in RViz.
10. Simulate robot movement and navigation in Gazebo/Webots environment for differential drive mobile robot applications.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Configure Ubuntu and ROS 2 environment using Virtual Box and dual boot techniques.
CO2	Demonstrate the use of Linux commands, ROS2 workspaces, packages.
CO3	Analyze the use of node creation for developing basic robotic programs.
CO4	Evaluate publisher–subscriber models in ROS2 for controlling and monitoring robot operation.
CO5	Design and visualize robot communication, TF frames, and robot states using RViz and rqt graphical tools.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2		2	2	3							2	3	1	2	1	1	1	3		1	
CO2	2	2	3		3							2	3	1	2	2	2	2	3			
CO3		2	3	2	3							2	3	1	2	1	1	2	3		1	
CO4			3	2	3			1	1	1		2	3		1			2	3		1	
CO5			3	2	3	1	1	1	1	1	2	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence

DATA MINING AND WAREHOUSING LAB

(27243105)

1. Data visualization and pattern analysis.
2. Creation of Star & Snowflake schema.
3. Implementation of OLAP operations .
4. Data preprocessing using Python/R.
5. Implementation of Decision Tree.
6. Naïve Bayes classification
7. Implementation of Apriori Algorithm.
8. Implementation of FP-Growth Algorithm.
9. Implementation of K-Means clustering.
10. Implementation of ETL process.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Describe and investigate the data.
CO2	Classify data to create various warehouse schemas.
CO3	Analyze different classification and clustering methods.
CO4	Evaluate association rules for pattern analysis.
CO5	Design storage and retrieval methods across various data sources.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2	3			2							2	3	2	2	2	2		2			
CO2	2	3	2		3							2	3	2	3	2	2	1	3			
CO3		2	3	1	3							2	3	1	2	1	1	2	3			
CO4		2	2		3			1	1	1	2	2	3	1	2	1	1	1	3		1	
CO5			3		3	1	1	1	1	1	3	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence

DEEP LEARNING LAB

(28243105)

List of Programs/ Experiments

1. Train a Deep learning model to classify a given image using pre trained model
2. Object detection using Convolution Neural Network
3. Recommendation system from sales data using Deep Learning
4. Improve the Deep learning model by tuning hyper parameters
5. Perform Sentiment Analysis in network graph using RNN
6. Image generation using GAN
7. Develop a model by using AUTO ENCODERS.
8. Implement an LSTM based Autoencoder in TensorFlow/Keras.
9. Perform Sentiment Analysis using RNN
10. Implement a pre-trained model on Keras for Transfer Learning

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Recognize the characteristics of deep learning models that are useful to solve real-world problems.
CO2	Implement Convolution Neural Network for solving various problems pertaining to image processing.
CO3	Analyze an end-to-end deep learning workflow.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	2		3		3							3	3	1	2	1	1	2	3			
CO2		3	3	2	3			1	1	1		3	3	1	2	2	2	2	3		1	
CO3			3	2	3	1	1	1	1	1	3	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence

CLOUD COMPUTING & VIRTUALIZATION LAB

(31243106/ 27243106)

1. Install Virtualbox/VMware Workstation with different flavors of linux or windows OS on top of windows 7 or 8.
2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs.
3. Install Google App Engine. Create a hello world app and other simple web applications using python/java.
4. Use GAE launcher to launch the web applications.
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6. Implement a procedure to transfer the files from one virtual machine to another virtual machine.
7. Implement a procedure to launch virtual machine using trystack (Online Openstack)
8. Install Hadoop single node cluster and run simple applications like word count.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Configure virtualization software.
CO2	Develop and run programs in a virtualized environment.
CO3	Design web applications using Python/ Java, employing GAE Launcher.
CO4	Evaluate cloud scenarios using CloudSim.
CO5	Analyse resource allocation techniques for virtual environments.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1	1	2	1			1	2	1	1	1	2	2	2	2	1	2	1	1	
CO2	3	3	2	1	2		1		1	1	1	2	2	2	3	3	3	1	2		1	1
CO3	3	3	3	2	3	1			2	1	2	3	3	2	3	3	3	2	3	1	2	
CO4	3	3	3	3	3			1	1	1	2	3	3	2	3	3	3	2	3		2	
CO5	3	3	2	2	3	3	1	2	1	2	1	2	3	3	3	3	3	2	3	3	2	1

Centre for Artificial Intelligence

INTERNET OF THINGS LAB

(24243106)

List of Programs/ Experiments

1. Introduction to Arduino Board and Arduino IDE installation and setup.
2. Program to blink and fade an LED using digital and analog output.
3. Program to read analog input and display sensor values on Serial Monitor.
4. Program to interface push buttons and control LED operations.
5. Program to measure distance using Ultrasonic Sensor.
6. Program to measure temperature and humidity using DHT11 sensor.
7. Program to measure light intensity using Light Dependent Resistor (LDR) sensor.
8. Program to detect motion using Passive Infrared (PIR) sensor.
9. Program to detect fire using Flame Sensor and generate alerts.
10. Program to control electronic appliances using Relay Shield.
11. Program to generate melody using Piezo buzzer and display digits using Seven Segment Display.
12. Program to control Servo Motor movement and wireless servo control using Bluetooth.
13. Design and implementation of smart irrigation and weather monitoring systems using IoT sensors and cloud platforms.
14. Design and implementation of home automation and real-time monitoring systems using Blynk, Node-RED, ThingSpeak, ESP32, and Raspberry Pi.
15. Design and implementation of wireless communication and automation systems using XBee, LoRa, robotics, and industrial IoT applications.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Explain Arduino programming, sensor interfacing, and basic IoT hardware components.
CO2	Develop IoT-based monitoring and automation systems using sensors, actuators, and wireless communication technologies.
CO3	Analyze IoT platforms and tools such as Node-RED, Blynk, ThingSpeak, and ESP32 for real-time applications.
CO4	Implement smart IoT applications for automation, monitoring, and communication systems.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2	1									3	2	2	2	2	2					
CO2	3	3		2								3	2	2	2	3	3				1	
CO3	3	3	3	2	2							3	3	2	3	3	3	2	2		1	
CO4		2	3	2	3			1	1	1		3	3	1	2	1	1	2	3		1	

Centre for Artificial Intelligence

CLOUD COMPUTING & BIG DATA LAB

(28243106)

List of Programs

1. Install and configure a Type-2 hypervisor (Oracle VirtualBox/VMware) to create a local, isolated environment for running multiple operating systems and cloud tools
2. Set up a fully functional Linux development environment inside a VM by performing essential installations and configurations.
3. Learn to control a VM using command-line tools, which is crucial for managing headless servers in the cloud
4. Gain hands-on experience with Infrastructure-as-a-Service (IaaS) by launching a virtual server on a public cloud platform
5. Set up a single-node Hadoop cluster on a Linux VM, which includes configuring HDFS and YARN for local development
6. Execute fundamental file operations in HDFS, the primary storage system for Hadoop
7. Implement the classic "WordCount" example to understand the core MapReduce programming model, an industry-standard introduction to Hadoop analytics.
8. Solve a practical data-mining problem by writing a MapReduce program to find extreme values within a large dataset.
9. Implement matrix multiplication using MapReduce, a common foundational algorithm for more advanced data processing.
10. Configure a fully distributed Hadoop cluster across multiple VMs to understand its architecture and scalability.
11. Use Apache Pig's scripting language, Pig Latin, to process structured and semi-structured data, which simplifies complex dataflows.
12. Use Apache Hive to perform data analysis using HiveQL, a SQL-like language for querying data stored in HDFS.
13. Apply learned technologies to an open-ended real-world scenario using the Hadoop ecosystem
14. Use the CloudSim simulation framework to model and simulate cloud computing components and policies without real hardware
15. Get exposure to open-source cloud platforms (OpenStack) and modern application containerization (Docker/Kubernetes).

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Perform cloud environment setup and configuration.
CO2	Implement virtualization and cloud services.
CO3	Analyze hadoop ecosystem tools.
CO4	Evaluate big data applications.
CO5	Design practical cloud and big data solutions.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2			2					1		1	3	2	2	2	2		2			
CO2	2	3	2		3								3	2	3	2	2	1	3			
CO3		3	2		2	1							2	1	2	2	2	2	2	1		
CO4		2	3	2	3			1	1	1			3	1	2	1	1	2	3		1	
CO5		2	2		2	2	1	1	1	1	2		2	1	1	1	1	2	2	2	1	1

Centre for Artificial Intelligence

STATISTICAL COMPUTING LAB

(31243108)

List of Programs/ Experiments

1. Build preprocessing pipelines for real datasets.
2. Implement automated missing-value and outlier handling systems.
3. Compare execution time of loop-based and vectorized statistical operations.
4. Create statistical visualizations for multidimensional datasets.
5. Program computational hypothesis testing routines.
6. Simulate sampling distributions using repeated random sampling.
7. Implement Monte Carlo simulation for probabilistic estimation.
8. Build regression models and analyze residual errors.
9. Implement gradient descent for parameter optimization.
10. Perform bootstrapping and permutation-based inference.
11. Create mini statistical dashboards using computational visualization.
12. Analyze computational efficiency for large datasets.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Design efficient data preprocessing pipelines using vectorized operations
CO2	Implement computational algorithms to perform non-parametric inference, hypothesis testing, and gradient-descent-based regression modeling.
CO3	Create interactive statistical dashboards and multidimensional visualizations

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**WK1**	**WK2**	**WK3**	**WK4**	**WK5**	**WK6**	**WK7**	**WK8**	**WK9**
CO1	3	3	3	2	3			1	1	1	1	3	3	2	3	3	3	2	3		2	
CO2	3	3	3	3	3	1	1	1	1	1	1	3	3	2	3	3	3	2	3	1	2	1
CO3	3	3	2	3	3	1	1	1	2	2	2	3	3	2	3	3	3	2	3	1	3	1

Centre for Artificial Intelligence

DATA SCIENCE LAB

(24243108)

List of Programs/Experiments

1. Basic Python syntax, variables, data types, control structures, functions.
2. Working with NumPy: Array creation, indexing, slicing, reshaping. Vectorized operations and broadcasting.
3. Data Manipulation using Pandas: Creating DataFrames and Series. Handling missing values, filtering, sorting, merging, grouping.
4. Data Import and Export: Loading CSV, Excel, JSON, and SQL data. Writing datasets to files.
5. Exploratory Data Analysis (EDA) – Descriptive Statistics: Summary statistics, distributions, cross-tabulation. Detecting outliers.
6. Data Visualization using Matplotlib: Line, bar, histogram, scatter, box plots. Subplots and styling.
7. Data Visualization using Seaborn: Pairplot, heatmap, countplot, distplot. Correlation matrix visualization.
8. Data Cleaning and Preprocessing: Handling missing data, encoding, scaling, normalization. Date-time processing.
9. Feature Engineering: Creating new features. Binning, one-hot encoding, label encoding. Dimensionality reduction basics (PCA).
10. Linear Regression: Implement simple and multiple linear regression. Visualize regression lines. Evaluate using RMSE, MAE, R^2.
11. Logistic Regression: Binary classification using logistic regression. Confusion matrix, accuracy, precision, recall, F1-score.
12. Decision Trees & Random Forests: Building decision tree classifiers/regressors. Applying ensemble learning (Random Forest). Feature importance analysis.
13. K-Means Clustering: Implement K-Means. Elbow method for optimal K. Cluster visualization.
14. Naïve Bayes Classifier: Text classification (spam detection or sentiment analysis).
15. Model Evaluation & Validation: Train-test split, k-fold cross-validation. Hyperparameter tuning with GridSearchCV.
16. Time Series Analysis (Basic): Plotting time series. Moving averages, decomposition.

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Apply Python programming and essential data manipulation techniques.
CO2	Implement exploratory data analysis (EDA) and visualize datasets.
CO3	Evaluate machine learning models.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2										3	2	2	2	2	2					
CO2	3	3		2	2							3	3	2	3	3	3		2		1	
CO3	2	3		3	2							3	3	2	2	2	2		2		2	
CO4	2	3	2		3			1	1	1		3	3	2	3	2	2	1	3			
CO5		2	3		3	1	1	1	1	1	2	3	3	1	2	1	1	2	3	1	1	1

Centre for Artificial Intelligence
IMAGE PROCESSING LAB
(28243108)

1. **Read, Display, and Save an Image –** Load an image from disk, display it using a library (e.g., OpenCV, PIL), and save it in a different format.
2. **Color Space Conversion –** Convert an RGB image to grayscale, HSV, and CMYK; display each channel separately.
3. **Geometric Transformations –** Apply translation, rotation, scaling, and affine (shear) transforms to an image.
4. **Image Arithmetic –** Add, subtract, multiply, and divide two images (e.g., blend two pictures or compute image differences).
5. **Histogram Equalization –** Compute and display the intensity histogram of a grayscale image, then apply histogram equalization to improve contrast.
6. **Noise Addition and Smoothing –** Add Gaussian and salt-and-pepper noise to an image, then reduce noise using mean, median, and Gaussian filters.
7. **Edge Detection –** Implement or apply Sobel, Prewitt, and Canny edge detectors; compare their outputs.
8. **Image Thresholding –** Convert a grayscale image into a binary image using global thresholding (Otsu's method) and adaptive thresholding.
9. **Morphological Operations –** Perform erosion, dilation, opening, and closing on a binary image (e.g., to remove small noise or close gaps).
10. **Image Sharpening –** Enhance edges using a Laplacian filter or unsharp masking.
11. **Frequency Domain Filtering –** Compute the Fourier transform of an image, apply a low-pass (blurring) and high-pass (edge enhancement) filter, then inverse transform.
12. **Image Segmentation using K-Means –** Cluster pixels by color/intensity using the k-means algorithm to segment the image into distinct regions.
13. **Template Matching –** Find a small template image inside a larger image using correlation or normalized cross-correlation.
14. **Image Rotation without Built-in Functions –** Write your own rotation algorithm (using nearest-neighbor or bilinear interpolation) without using library rotate functions.
15. **Video Background Subtraction –** From a static camera video, model the background and extract moving foreground objects (e.g., using frame differencing or MOG2).

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Perform image acquisition and basic image operations.
CO2	Implement image enhancement and filtering methods.
CO3	Analyze segmentation and edge detection techniques.
CO4	Evaluate image processing applications using software tools.
CO5	Develop image processing projects.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	2									2	3	3	2	2	2	2				1	
CO2	2	3		2								3	3	2	2	2	2				1	
CO3	2	3		3	2							3	3	2	2	2	2		2		2	
CO4		2	2	3	3			1	1	1		3	3	1	2	1	1	1	3		2	
CO5			3	2	3	1	1	1	1	1	3	3	3		1			2	3	1	2	1

Centre for Artificial Intelligence
CORNERSTONE PROJECT
(31243109/ 24243109/ 27243109/ 28243109)

1. Design and Development of an Intelligent Vision-Based Human Activity Recognition Framework Using Deep Learning
2. Development of a Deep Neural Network-Based Framework for Multi-Modal Medical Image Enhancement and Analysis
3. Design of an Energy-Efficient Approximate Square Root and CORDIC Architecture for Edge DSP Applications
4. Development of an AI-Driven Intrusion Awareness and Threat Intelligence Framework for Smart Networks
5. Design and Development of an Intelligent Traffic Management and Vehicle Analytics System Using Computer Vision
6. Development of an Edge-AI Enabled Real-Time Plant Disease Diagnosis Framework Using Lightweight MobileNet Architectures
7. Design and Development of an Intelligent Network Traffic Classification System using Machine Learning.
8. Development of an Intelligent Chatbot Using Natural Language Processing
9. Design and Implement Quantum Secure Encryption Schemes
10. Design and Development of an AI-Based Plant Disease Detection System
11. Development of a Unified Intelligent Learning Framework Integrating ELM, SVM, BLS, and RVFL Networks for Imbalanced Data Classification
12. Design and Development of a Deep Learning-Driven Predictive Control Framework for Nonlinear Dynamical Systems
13. Development of an Intelligent Multilingual Neural Machine Translation System Using Artificial Intelligence
14. Design and Development of a Novel Cross-Modality Image Fusion and Registration Framework for Misaligned Image Analysis
15. Development of an Intelligent Automatic Text Summarization Framework Using Transformer-Based Deep Learning Models
16. Design and Development of an AI-Driven Intelligent Supply Chain Management and Decision Support Framework
17. Development of an Intelligent AI-Based Fault Detection, Diagnosis, and Classification Framework for Industrial Systems
18. Design and Development of an AI-Powered Cyberbullying Detection and Prevention Framework for Social Media Platforms
19. Design and Development of Voice Controlled AI Application Assistant.
20. Development of Real-Time Intrusion Awareness and Detection System Using Machine Learning
21. Design an Adaptive Deep Learning Framework for Robust Low-Light Image Enhancement and Restoration
22. Design of an Energy-Efficient Approximate Multiplier Architecture for Edge-AI CNN Accelerators
23. Development of an Intelligent Intrusion Detection and Cyber Defense Framework for Next-Generation Cybersecurity Systems
24. Develop an intelligent Real-Time Face Mask Compliance Monitoring Using Deep Learning and Computer Vision
25. Development of a hybrid Vision Transformer and Diffusion Model Framework for Accurate Plant Disease Detection and Classification
26. Design and Development of an AI-Based Intrusion Detection System for Secure Computer Networks.
27. Design and Implementation of a Fake News Detection System Using Machine Learning
28. Design and Implement Quantum Secure Digital signatures
29. Design and Development of a Vision-Based Driver Drowsiness Detection System
30. Development of an adaptive Neural Control Strategies for Uncertain Nonlinear Dynamical Systems under External Disturbances
31. Development of a Hybrid Broad Learning Framework for Imbalanced Data Analytics
32. Design of an Emotionally Intelligent Conversational Agent Using Deep Learning and NLP
33. Development of a Multiscale State Space Learning Framework for Multisource Remote Sensing Classification
34. Design of a Transformer-Based Intelligent Sentiment Analytics Platform
35. Development of an AI-Driven Sustainable Green Supply Chain Optimization System
36. Design of a Smart AI-Enabled Power Grid Management and Decision Support Framework
37. Development of an Explainable AI Framework for Fake Profile Detection in Online Social Networks
38. Design and Development of vision based Robotic controlled Manipulators.
39. Development of a Deep Learning-Based Retinal Image Enhancement for Improved Disease Diagnosis
40. Development of Hybrid Intrusion Awareness and Prevention System Using Deep Learning
41. Development of a Human Activity Recognition and Fall Detection using Vision-Based AI Techniques
42. Hybrid Deep Learning and GAN-Based Framework for Plant Disease Classification - Technology: CNN + GAN - Contribution: Improved classification using synthetic data
43. Design and Development of Smart Routing Optimization using Reinforcement Learning in Computer Networks.

44. Implementation of Credit Card Fraud Detection Using Machine Learning Algorithms
45. Design and Implement Zero knowledge Protocols based Signatures
46. Imbalanced Data Classification Using ELM, SVM, BLS, and RVFL Networks
47. Design and Development of a Computer Vision-Based Smart Attendance System
48. Design and Development of a Residual-Driven Intelligent Framework for Cyber Attack Detection and Resilient Nonlinear Control
49. Development of an AI-Powered Clinical Decision Support Framework for Automated Medical Image Diagnosis
50. Design and Development of a Lightweight Feature Fusion Network for Accurate Visual Change Detection
51. Development of a Deep Learning Framework for Robust Speech Emotion Recognition
52. Design and Development of a Fuzzy Logic-Based Intelligent Inventory Optimization and Decision Support Framework
53. Development of a Machine Learning Framework for Predictive Condition Monitoring and Maintenance Analytics
54. Design and Development of a Deep Learning-Based Framework for Emotion Analytics in Social Media Interactions
55. Development of an AI-Enabled Multimodal Biometric Authentication Framework for Secure Identity Management
56. Design and Development of a Transformer-Based Satellite Image Restoration Framework for Enhanced Remote Sensing Applications
57. Design and Development of a High-Performance Approximate Neural Computing Accelerator for Edge-AI Applications
58. Development of a Blockchain-Assisted Cyber Threat Monitoring and Intrusion Awareness Framework
59. Design and Development of a Deep Learning Framework for Image Forgery and Synthetic Image Detection
60. Development of an AI-Driven Traffic Engineering Framework for Software Defined Networks
61. Design and Development of a Multimodal Deep Learning Framework for Human Emotion Recognition
62. Development of a Knowledge-Guided Transformer Framework for Advanced Object Detection
63. Design and Development of an AI-Powered Cyber Attack Detection and Autonomous Response System
64. Development of a Social Network Intelligence Framework for Influential User Discovery and Community Analytics
65. Design and Development of a ROS-Based Autonomous Robotic Manipulation Framework for Intelligent Industrial Applications
66. Development of an Intelligent Deep Learning Framework for Underwater Image Restoration and Enhancement
67. Design and Development of a Cyber Threat Intelligence-Driven Framework for Proactive Intrusion Detection and Response
68. Development of an AI-Based Smart Waste Segregation System for Sustainable Waste Management
69. Design and Development of a Deep Learning Framework for Network Congestion Prediction and Adaptive Traffic Optimization

COURSE OUTCOMES

After completion of this course, the students would be able to:

CO1	Apply Artificial Intelligence and Data Science techniques to analyze, preprocess, and model complex datasets for intelligent decision-making and problem-solving.
CO2	Demonstrate conceptual understanding of machine learning algorithms, statistical analysis, and AI-driven methodologies for developing data-centric engineering solutions.
CO3	Integrate mathematical modeling, computational intelligence, and domain knowledge to design efficient, scalable, and innovative AI-based applications.
CO4	Utilize modern AI, machine learning, and analytical tools to evaluate model performance, optimize accuracy, and validate computational results effectively.
CO5	Interpret analytical outcomes critically and recommend practical, sustainable, and ethically responsible solutions for real-world engineering and societal challenges.

CO-PO Mapping Matrix														CO-WK Mapping Matrix								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1	3	3	2	2	3	-	-	-	-	-	2	3	3	2	3	3	3	1	3		2	
CO2	3	3	2	2	2	-	-	-	-	-	2	3	2	2	3	3	3	1	2		2	
CO3	3	3	3	2	3	-	-	-	2	-	2	3	3	2	3	3	3	2	3		2	
CO4	3	3	2	3	3	-	-	-	2	-	2	3	3	2	3	3	3	1	3		3	
CO5	2	3	2	2	2	2	3	3	2	2	3	2	3	2	2	3	3	2	2	2	3	3